IDENTITY

Invited reviewer
B2B guest in the tenant

Entra ID app
Single-tenant, assignment required

Assignment required
Unassigned users get no token

Uninvited blocked
No guest invite, no session

Local run uses a dev user, not Entra. Hosted demo is invite-only. Managed identity is the app credential. No access keys in images.

RESOURCE GROUP rg-mps-demo

Container Apps environment — network boundary

Simulator
Three plants, one seed

Static Web App
Harness over console

API
Sites routes. Scale 0 to 3.
Escalated approve is refused

Worker
Scales on queue depth

Mock enterprise
Quality, planning, maintenance
Scenario evidence on read

Queue (demo: in-process)
API hands off the investigation
Production: Azure Service Bus

Store (demo: SQLite)
Incidents, audit, spend ledger
Production: Azure SQL

Index (demo: in-process)
Documents and records
Production: Azure AI Search

Azure OpenAI
gpt-5.4-mini, TPM quota
Production: PTU or quota tier

Key Vault
No secrets in the image

Managed identity
RBAC to SQL, OpenAI, Search, Vault

App Insights
OTel GenAI traces and alerts

Spend cap
$2 per day, then replay

Production swaps: Service Bus for the queue, Azure SQL for SQLite, Azure AI Search for the in-process index, Front Door with WAF, private endpoints, and APIM in front of the model gateway. Rollback is a Container Apps revision.

Northwater Fab
Simulated front-end fab
SC-5 lot starts here

Ironwood Fab
Simulated front-end fab
Same SC-5 lot

Lumen Pilot Works
Simulated strip and pilot
Not on that lot

TARGETS

API p95
under 300 ms
Investigations are async

Investigation p95
under 45 s
Tools plus two model calls

Unit cost
$0.02 to $0.10
Per successful investigation

Hard cap
$2 per day
Over cap switches to replay

Scale
API 0-3
Worker on queue. Simulator x1

Not this drawing
docker compose
API, mocks, web, Aspire

Talk track only
Front Door + WAF
Private endpoints, Event Hubs

GITHUB ACTIONS — ci.yml does not deploy

ci.yml
Build, test, coverage, format, CodeQL, gitleaks, dependency review, Bicep build

deploy.yml
Manual only. Environment azure-demo. Required reviewer. Bicep what-if, then deploy.

OIDC to Azure
No stored cloud secrets. Push images to ACR. New Container Apps revision.

LEGEND

- Identity. Entra, invite only.
- Resource group, harness, and console.
- Targets, scale, and the spend cap.
- Traces, Key Vault, and the simulated plants.
- Manual deploy gate.
- OIDC. No stored cloud secrets.
- Data nodes: demo service, then the production swap.

MERIDIAN PRECISION SYSTEMS
Quality and Engineering Issue Resolution

Physical architecture
G2 · Issue #47 Version v1.2 Date 2026-10-14
Prepared by StatusNeo

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